

```

u8 temp_count;
u32 temp;
void temp_process()
{
    if(temp_count>=100)
    {
        temp_count=0;
        temp=(u32)(temp_read()*10000);
    }
}
union AT24C02_u16
{
    u16 a;
    u8 b[2];
}u16_write,u16_read;
union AT24C02_s16
{
    s16 a;
    u8 b[2];
}s16_write,s16_read;
union AT24C02_float
{
    float a;
    u8 b[4];
}float_write,float_read;
union AT24C02_u32
{
    u32 a;
    u8 b[4];
}u32_write,u32_read;
u8 string_write[11]={"HELLO WORLD"};
u8 string_read[11];
void AT24C02_process()
{
    u8 i=0;

    u16_write.a=1234;
    for(i=0;i<sizeof(u16);i++)
        AT24C02_write(0x30+i,u16_write.b[i]);
    for(i=0;i<sizeof(u16);i++)
        u16_read.b[i]=AT24C02_read(0x30+i);
    s16_write.a=-1234;
    for(i=0;i<sizeof(s16);i++)
        AT24C02_write(0x20+i,s16_write.b[i]);
    for(i=0;i<sizeof(s16);i++)
        s16_read.b[i]=AT24C02_read(0x20+i);
    float_write.a=3.1415926;
    for(i=0;i<sizeof(float);i++)
        AT24C02_write(0x10+i,float_write.b[i]);
    for(i=0;i<sizeof(float);i++)

```

```

        float_read.b[i]=AT24C02_read(0x10+i);
    u32_write.a=70000;
    for(i=0;i<sizeof(u32);i++)
        AT24C02_write(0x00+i,u32_write.b[i]);
    for(i=0;i<sizeof(u32);i++)
        u32_read.b[i]=AT24C02_read(0x00+i);
    for(i=0;i<sizeof(string_write);i++)
        AT24C02_write(0x40+i,string_write[i]);
    for(i=0;i<sizeof(string_write);i++)
        string_read[i]=AT24C02_read(0x40+i);
}
u8 AD_count;
u8 ch0,ch1,ch3;
u16 ch3_05;
u8 ch3_0_99,ch3_1_5;
void AD_process()
{
    if(AD_count>=10)
    {
        AD_count=0;
        AD_read(0x41);
        ch1=AD_read(0x41);
        AD_read(0x43);
        ch3=AD_read(0x43);

        ch3_05=ch3*100/51;
        ch3_0_99=ch3/2.57;
        ch3_1_5=ch3/51.1+1;
    }
}
u8 DA_count;
void DA_process()
{
    static u8 haha;
    if(DA_count>=2)
    {
        DA_count=0;
        DA_write(haha++);
    }
}
u8 hour,minute,second,year,month,day,date;
void DS1302_process()
{
    second = BCDToDec(Read_Ds1302_Byte(0x81));
    minute = BCDToDec(Read_Ds1302_Byte(0x83));
    hour = BCDToDec(Read_Ds1302_Byte(0x85));
    year = BCDToDec(Read_Ds1302_Byte(0x8d));
    month = BCDToDec(Read_Ds1302_Byte(0x89));
    day = BCDToDec(Read_Ds1302_Byte(0x8b));
    date = BCDToDec(Read_Ds1302_Byte(0x87));

```

```

}
u16 NE555_count;
u16 NE555_date;
void NE555_process()
{
    if(NE555_count>=1000)
    {
        NE555_count=0;
        NE555_date=(TH0<<8)|TL0;
        TL0=0;
        TH0=0;
    }
}
u8 CSB_count;
u16 CSB_date;
void CSB_process()
{
    if(CSB_count>=100)
    {
        CSB_count=0;
        CSB_date=CSB_get();
    }
}
void CK_process()
{
    if(CK_flag)
    {
        drive(CK_string[0],0x80);
        drive(CK_string[1],0x80);
        drive(CK_string[2],0x80);
        CK_flag=0;
    }
}
u8 pwm_count=1;
u8 compare = 90;
void PWM_prcoess()
{
    pwm_count=pwm_count%10+1;
    if(pwm_count<=(compare/10))
        drive(0xff,0x80);
    else
        drive(0x00,0x80);
}
u8 led_count;
void led_process()
{
    if(led_count>=10)
    {
        led_count=0;

```

```

    }
}
void main()
{
    system_init();
    Timer2_Init();
    Counter0_Init();
    Uart1_Init();
    temp_read();
    delay_ms(750);
    DS1302_set(11,25,25,24,2,12,1);
    AT24C02_process();
    while(1)
    {
        SMG_set();
        key4_process();
        key16_process();
        temp_process();
        AD_process();
        DA_process();
        DS1302_process();
        CSB_process();
        CK_process();
    }
}
void Timer2_Isr(void) interrupt 12
{
    static u16 count;
    count++;
    key4_count++;
    key16_count++;
    temp_count++;
    AD_count++;
    DA_count++;
    NE555_count++;
    CSB_count++;
    led_count++;

    SMG_display();
    led_process();
    NE555_process();
    PWM_prcoess();
    ck_relax();
}

void Counter0_Init(void)
{
    TMOD|=0x05;
    TL0=0x00;

```

```

    TH0=0x00;
    TR0=1;
}
bit busy;
bit CK_flag;
u8 ck_time;
u8 CK_count;
u8 CK_string[3];
void ck_relax()
{
    ck_time++;
    if(ck_time>=50)
    {
        ck_time=0;
        CK_count=0;
        memset(CK_string,'\0',sizeof(CK_string));
    }
}
void Uart1() interrupt 4
{
    if (RI)
    {
        RI = 0;
        ck_time=0;
        CK_string[CK_count++] = SBUF;
        if(CK_string[CK_count-1]!='\n')
        {
            CK_count=0;
            CK_flag=1;
        }
    }
    if (TI)
    {
        TI = 0;
        busy = 0;
    }
}
void SendData(u8 dat)
{
    while (busy);
    busy = 1;
    SBUF = dat;
}
void SendString(char *s)
{
    while (*s)
    {
        SendData(*s++);
    }
}

```

```

sbit DQ=P1^4;
float temp_read()
{
    float temp;
    u8 low,high;
    init_ds18b20();
    Write_DS18B20(0xcc);
    Write_DS18B20(0x44);
    init_ds18b20();
    Write_DS18B20(0xcc);
    Write_DS18B20(0xbe);
    low=Read_DS18B20();
    high=Read_DS18B20();
    temp=(high<<8|low)*0.0625;
    return temp;
}
sbit sda=P2^1;
sbit scl=P2^0;
void AT24C02_write(u8 add, u8 dat)
{
    I2CStart();
    I2CSendByte(0xa0);
    I2CWaitAck();
    I2CSendByte(add);
    I2CWaitAck();
    I2CSendByte(dat);
    I2CWaitAck();
    I2CStop();
    delay_ms(5);
}
u8 AT24C02_read(u8 add)
{
    u8 dat;
    I2CStart();
    I2CSendByte(0xa0);
    I2CWaitAck();
    I2CSendByte(add);
    I2CWaitAck();
    I2CStart();
    I2CSendByte(0xa1);
    I2CWaitAck();
    dat=I2CReceiveByte();
    I2CSendAck(1);
    I2CStop();

    return dat;
}
void DA_write(u8 dat)
{
    I2CStart();

```

```

    I2CSendByte(0x90);
    I2CWaitAck();
    I2CSendByte(0x40);
    I2CWaitAck();
    I2CSendByte(dat);
    I2CWaitAck();
    I2CStop();
}
u8 AD_read(u8 add)
{
    u8 dat;
    I2CStart();
    I2CSendByte(0x90);
    I2CWaitAck();
    I2CSendByte(add);
    I2CWaitAck();
    I2CStart();
    I2CSendByte(0x91);
    I2CWaitAck();
    dat=I2CReceiveByte();
    I2CSendAck(1);
    I2CStop();
    return dat;
}
sbit SCK=P1^7;
sbit SDA=P2^3;
sbit RST=P1^3;
#define DecToBCD(dec) (dec/10*16)+(dec%10)
#define BCDToDec(bcd) (bcd/16*10)+(bcd%16)
void DS1302_set(u8 hour,u8 minute,u8 second,u8 year,u8
month,u8 day,u8 date)
{
    Write_Ds1302_Byte(0x8e,0x00);
    Write_Ds1302_Byte(0x80,DecToBCD(second));
    Write_Ds1302_Byte(0x82,DecToBCD(minute));
    Write_Ds1302_Byte(0x84,DecToBCD(hour));
    Write_Ds1302_Byte(0x86,DecToBCD(date));
    Write_Ds1302_Byte(0x88,DecToBCD(month));
    Write_Ds1302_Byte(0x8a,DecToBCD(day));
    Write_Ds1302_Byte(0x8c,DecToBCD(year));
    Write_Ds1302_Byte(0x8e,0x80);
}
void CSB_send()
{
    u8 i;
    for(i=0;i<2;i++)        //发送 2 个 40khz 方波
    {
        P10=1;
        delay12us();
        P10=0;

```

```

        delay12us();
    }
}
u16 CSB_get()
{
    u16 distance;
    CSB_send();
    TH1=0;TL1=0;
    TR1=1;
    while((P11==1)&&(TF1==0));
    TF1=0;
    TR1=0;
    distance=(TH1<<8|TL1)*0.017;
    return distance;
}
//u16 CSB_get()
//{
//    u16 distance;
//    CMOD=0x00;
//    CH=CL=0;
//    CSB_send();
//    CR=1;
//    while((P11==1)&&(CF==0));
//    CF=0;
//    CR=0;
//    distance=(CH<<8|CL)*0.017;
//    return distance;
//}
u8 key16_io()
{
    u8 key_io=0xff;
    P3=0xf0;P4=0xff;
    if(P44==0)    key_io=0x70;
    if(P42==0)    key_io=0xb0;
    if(P35==0)    key_io=0xd0;
    if(P34==0)    key_io=0xe0;
    P3=0x0f;P4=0x00;
    if(P33==0)    key_io=key_io|0x07;
    if(P32==0)    key_io=key_io|0x0b;
    if(P31==0)    key_io=key_io|0x0d;
    if(P30==0)    key_io=key_io|0x0e;
    return key_io;
}
u8 trg16,cont16;
void key16()
{
    u8 date=key16_io()^0xff;
    trg16=date&(date^cont16);
    cont16=date;
}

```